M3 HACKATHON · RIYADH

EcoPulse Riyadh

Which roofs should Riyadh cool first, and who should pay for it?

Razi Moayed Money · Industrial Engineering, Prince Sultan University

THE PROBLEM

Riyadh is getting hotter, and most of our home power goes to fighting it

54.8 °C

Peak land surface temperature in 2022, up from 51.0 °C in 2018

455 km²

Heat-island area in 2022, about 43 km² bigger than in 2018

~70%

Share of home electricity used by air conditioning in summer

Green Riyadh is planting millions of trees, but a 2023 satellite study found it hasn't measurably cooled the city yet. Trees take time. Roofs can act faster.

Sources: Imam 2023 (Landsat 2018–2022); Alshahrani & Boait 2019, Energies

WHO I BUILT IT FOR

The people who decide have no quick, honest way to compare options

City planners

Need to compare heat, energy, carbon and cost across neighbourhoods before they ask for a budget.

Building owners

Want to know if a roof upgrade will actually pay back, and how long it takes.

The gap

The research exists, but it's scattered across papers, and most tools never say where their numbers come from.

Still to validate: I haven't interviewed planners or owners yet. That's step one of my roadmap.

WHAT I FOUND

In Riyadh, paint beats plants

10% vs 7.5%

Cooling energy cut by a cool roof vs a green roof in a hot, dry climate

~6× cheaper

Upfront cost per m², by my estimate (60 vs 350 SAR)

No water

Green roofs need irrigation in a city that runs on desalinated water

The catch: at today's 0.18 SAR/kWh, even a cool roof takes **~28 years** to pay back for the owner. That's a pricing problem, not an engineering one. The fix is policy.

Sources: Elnabawi & Saber 2023; SEC tariff. Costs are my estimates, editable in the tool.

THE SOLUTION

EcoPulse answers one question in four clicks

1

Where?

Industrial areas, apartments, housing or villas

2

Paint or plants?

White reflective paint, or plants on the roof

3

How many roofs?

One neighbourhood, or all of Riyadh

4

Who pays?

Slide the government's share and watch the owner's wait shrink

You get one plain sentence, a verdict, and what it means for Saudi Arabia. Every number shows its source.

Live demo: EcoPulse Riyadh prototype · built with the aid of AI

IMPACT SNAPSHOT

Cool roofs on 30% of homes in one km² of medium-density Riyadh

1.26M kWh
electricity saved per year

~717 t
CO_2 avoided per year

−0.24 °C
street air temperature, approx.

6.3M SAR
upfront cost (my estimate)

~28 yrs
payback at 0.18 SAR/kWh

0.50 SAR
per kWh needed for a 10-year payback

For Saudi Arabia: across all 455.6 km² of Riyadh's hot zones, the same plan cuts 327,000 t of CO_2 a year, 0.12% of the Saudi Green Initiative goal. If the city pays half, owners wait 14 years, not 28.

Linear scaling of published simulations. Cooling use (120 kWh/m²/yr), roof share and costs are my assumptions.

NEXT 90 DAYS

Validate, then map, then partner

30 DAYS

Validate

- Interview 3 planners and 10 owners
- Replace my cooling-use guess with real bills from 20 homes
- Get 3 local contractor quotes

60 DAYS

Map and test

- Add a Landsat heat map of Riyadh districts
- Coat one test roof, measure before and after
- Arabic interface

90 DAYS

Partner

- Present results to Riyadh municipality and Green Riyadh
- Add cool pavements and tree shade
- Publish the method openly

Help me test EcoPulse with real Riyadh data

Try it: engrazimoney.github.io/ecopulse-riyadh

A planner

to try the tool on a real district

Real bills

to replace my assumptions with data

A mentor

in urban climate or building energy

Razi Moayed Money · razimoney007@gmail.com · Built with the aid of AI · Idea first entered at NextStep Hacks 2026; prototype built for M3